
GARP-Blind Perturbations Outperform Minimal GARP Repair Under an Exogenous Payoff

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Abstract

Repairing an AI agent’s incoherent preferences is an established line of work: inference-time transitivity repairs, isotonic projections onto preference-defined cones, and training-time rationality penalties have all been proposed, several with reported downstream gains. No published result includes a displacement-matched control, so none establishes whether the reported gains come from restoring coherence or from displacement toward an interior point. We build that control. Given an LLM agent’s own choices over budget sets, we compute the minimal MILP perturbation restoring GARP-consistency, then compare it against a GARP-blind null operator spending the identical displacement budget shrinking every bundle toward a fixed interior point. Under the first payoff, the null outperforms the real repair by more than $2\times$ (mean Δpayoff 0.0220 vs. 0.0091, Wilcoxon $p = 3.98 \times 10^{-10}$, winning 81% of traces). A second payoff built to remove the first one’s exploitable fixed target reproduces the same verdict (null 0.0217 vs. real 0.0072, $p = 1.24 \times 10^{-5}$, 70.8% win rate). Trace extremity predicts the null’s advantage in both experiments. The paper’s contributions are a capacity-deconfounded identification strategy for the coherence–competence question and a discard-selection instrument-validity finding. That question remains open.

1 Introduction

An LLM agent asked to allocate a fixed budget across two goods at varying prices, repeatedly, will frequently violate the Generalized Axiom of Revealed Preference (GARP; Afriat, 1967, Varian, 1982): it prefers bundle x_1 to x_2 at one budget line and the reverse at another, in a way no monotone utility function can rationalize. The same budget-allocation instrument is the standard paradigm for testing GARP experimentally on human subjects [Andreoni and Miller, 2002, Choi et al., 2007, 2014], and a growing literature applies it to GPT-family models with no repair step, asking whether they satisfy classical rationality axioms [Chen et al., 2023]. A separate literature repairs this kind of inconsistency in LLM systems — restoring transitivity in pairwise judgments, projecting cardinal scores onto monotone cones, and penalizing incoherence during training — and several of these systems report that the repaired outputs are downstream-better than the raw ones. None of them establishes a *dose*: how much better, as a function of how much repair was needed, isolated from the capacity and training confounds that come from comparing a richer model class to a poorer one.

An LLM agent makes a sequence of budget-constrained choices over two goods. We compute, as a single mixed-integer linear program, the minimal quantity perturbation that would make the agent’s own realized choice sequence GARP-consistent, applied post hoc to a frozen agent: nothing about its weights, training, or decoding changes across doses. The size of that perturbation is a graded, cardinal dose. We score both the raw and the repaired sequence against a fixed, equal-weight Cobb–Douglas payoff whose optimum at each budget line is a closed-form function of prices and income alone —

never of the agent’s own choices — so the outcome measure is exogenous to the preference data by construction.

The design does not identify the effect of coherence restoration. An operator that spends the identical displacement budget shrinking every bundle toward a fixed interior point, with no access to the GARP violation structure, outperforms the real GARP-restoring repair on the exogenous payoff ($p = 3.98 \times 10^{-10}$; §5.1). The result holds at both model scales and under two payoff designs. No published axiom-enforcement result we are aware of includes this control. Restoring GARP-consistency buys no more exogenous payoff than displacement toward the interior of the budget line (0.0091 vs. 0.0220). We label this claim C1; C2 is the coherence–competence question and C3 the claim that the reciprocal-framing manipulation disrupts the agent.

The conjunction is new: an agent’s own realized choices, a graded coherence-indexed dose, and an outcome that is not itself a preference judgment, run together so that dose is not confounded with model capacity or training. A 100-session pilot (2 models $\times$ 2 framings $\times$ 25 seeds) preceded the main experiment. We run this at a model scale where the pilot found coherence headroom (1.5B parameters), and repeat the same pipeline at 3B, where the pilot found mean baseline CCEI (Critical Cost Efficiency Index; Afriat, 1972, 1973) 0.99. The 3B run is an identification control on the method. The two scales are not two points on a shared dose curve.

The reciprocal-price framing manipulation, intended to induce coherence variation at 1.5B, produced in the pilot a GARP pass-rate collapse from 0.76 to 0.17 at 3B under the same manipulation. The pilot’s separate CCEI estimate at 1.5B (+0.0169, $p = 0.66$) was never statistically significant, and the main experiment’s retry-corrected estimate at 1.5B (−0.0109, $p = 0.73$) is likewise indistinguishable from zero (§5.2). The discard rate is the more reliable signal that the manipulation disrupted the 1.5B agent (our claim C3): it is measured without conditioning on completion, and the CCEI point estimate is not; 52% of the pilot’s reciprocal-framing sessions at 1.5B were silently discarded for failing the output-format contract. Replacing single-turn elicitation with 25 sequential calls at 1.5B drops the GARP pass rate from 0.40 to 0.10, with no discard confound in either condition. The drop runs opposite to an independently published finding in the same domain and model family at a larger scale [Wang et al., 2025a].

Contributions. (1) An application of Demuyneck and Rehbeck [2023]’s minimal-quantity-error MILP projection to an agent’s own realized choices, verified independently on every output, not by solver status. (2) A closed-form exogenous payoff, fixed before any data collection, checked for leakage and for the mechanical confound that larger violations have more room to improve. (3) A dose-matched, GARP-blind null-operator control, run against two exogenous payoff designs. The measured raw dose–response relationship does not show that restoring GARP-consistency buys the exogenous payoff gain: displacement toward the interior buys as much (§5.1). (4) An instrument-validity finding (discard-selection bias under a disruptive manipulation), corrected by a capped retry protocol and reported per cell. (5) The single-turn-vs-multi-turn elicitation-format manipulation above, which runs opposite to an independently published finding in the same domain and model family at a larger scale [Wang et al., 2025a].

2 Related Work

Four criteria jointly define the claim: an agent’s own choices, an exogenous payoff, a graded rather than binary dose, and a minimum-distance projection. Of nine published repair systems, none satisfies all four (Table 1).

POISE. Wang et al. [2026] project with pool-adjacent-violators onto a closed convex chain-monotone cone, and prove the edit lands weakly closer to a posited ground truth. The GARP-consistent set is a union of polyhedra, hence not convex, so the non-expansiveness of projection onto a closed convex set in L_2 does not transfer (§B.1); the minimum-distance priority is theirs. POISE projects a *teacher’s* offline labels. Three further inference-time repairs have a binary edit and no Afriat machinery: Chadwick et al. [2025], TrustJudge [Wang et al., 2025b], and CONSISTRE [Sun, 2026]. LLM-RankFusion [Zeng et al., 2024] has an exogenous payoff but declines a minimum-distance method. Buchanan and Foster [2026] alters the agent itself at training time and reports no downstream comparison. Aguiar and Kashaev [2026] is not an LLM agent and is evaluated on forecasting, one operating on synthetic-data-trained forecasts of human survey consumers’ choices, not an LLM agent’s own decisions.

Table 1: Nine published repair systems scored against the four criteria that define the claim. *Own choices*: the repaired object is a choice the agent made for itself from a budget set. *Exogenous payoff*: an outcome into which no preference judgment enters; *graded dose*: the intervention has a continuous magnitude, not an on/off setting; *min.-distance projection*: the edit is the solution to an explicit minimum-distance program.

	Own choices	Exogenous payoff	Graded dose	Min.-distance projection
POISE [Wang et al., 2026]	no	no	no	yes
Rationality layer [Chadwick et al., 2025]	no	no	no	yes
TrustJudge [Wang et al., 2025b]	no	no	no	no
CONSISTRE [Sun, 2026]	no	partial	no	no
LLM-RankFusion [Zeng et al., 2024]	no	yes	no	no (declined)
Innate preferences [Buchanan and Foster, 2026]	yes	no	no	no
GARP-EFM [Aguir and Kashaev, 2026]	no	yes	no	no
HRC/DSPPO [Huang et al., 2026]	no	no	yes	no
Control vectors [Cook et al., 2026]	yes	no	yes	no
This paper	yes	yes	yes	yes

91 **Preference-model variants.** A parallel line varies the transitivity of the *preference model* rather than
 92 an agent’s choices, confounding coherence with capacity by construction: GPM [Zhang et al., 2025]
 93 loses on its own length-controlled metric in 18 of 24 cells, and HRC/DSPPO’s [Huang et al., 2026]
 94 inverted-U dose–response curve is over a training-time schedule weight scored by an LLM judge, not
 95 an exogenous payoff. Ours acts post hoc on a payoff that is exogenous by construction. Andrews
 96 [2026] proposes 1 – CCEI as a training penalty on a priori grounds and reports no measurement of
 97 it; we report the measurement that proposal calls for.

98 **Psychometric reliability.** Nitsch et al. [2022] find that CCEI and Houtman–Maks [Houtman and
 99 Maks, 1985] test–retest reliability (ICC) ranges from 0.158 to 0.685 across eight datasets, never
 100 reaching the conventional “good” threshold of 0.75, and that mean CCEI fell among participants
 101 given a chance to revise their own choices. Their revision arm re-presented a random subset of trials
 102 with no consistency objective, so it could not guarantee fewer violations; ours attains rationalizability
 103 by construction. Repair by construction rather than by re-presentation also separates ours from Yamin
 104 et al. [2026]’s isotonic repair, which their own evaluation reports as worse in 14 of 16 cells. Nitsch
 105 et al. diagnose their finding as a between-subject-variance problem whose prescribed remedy, a
 106 between-groups design, is this paper’s design (§6 bounds how far our power supports it). Three
 107 independently published axiom-enforcement results — Nitsch et al. [2022], Yamin et al. [2026], and
 108 Zhu et al. [2025] — all report that enforcing axioms failed to improve the outcome measured. We
 109 therefore carry a null-operator control that none of the three had.

110 **Economic-choice steering.** Cook et al. [2026], in the economics literature, steer economic choices
 111 toward payoff-maximizing behavior with personas and control vectors. They meet two of the four
 112 criteria and use no Afriat machinery. They find that reframing moves models while persona prompting
 113 does not, which corroborates our manipulation choice. Wang et al. [2025a] find the same in the same
 114 domain and model family (Qwen2.5) at 7B, where collapsing multi-turn to single-turn drops CCEI
 115 by 0.241. Which format is more degrading reverses between their 7B model and ours: reversing their
 116 manipulation reverses the sign, and the GARP pass rate falls from 0.40 to 0.10 (§5.3).

117 3 Method

118 3.1 Minimal-quantity-error MILP projection

119 The MILP formulation below is Demuynck and Rehbeck [2023]’s multiplier-free ordinal characteri-
 120 zation of minimal-quantity-error GARP repair, applied here to an LLM agent’s own choice sequence.
 121 For a sequence of T observations $(p_t, x_t)_{t=1}^T$, G goods, GARP holds if there is no sequence of choices
 122 such that x_{t_1} is revealed preferred to $x_{t_2}, \dots, x_{t_n}$ is revealed preferred to x_{t_1} , with at least one strict
 123 revealed preference in the cycle. Given a GARP-violating sequence, we seek the minimal perturbation
 124 $\tilde{x}_t$ that restores GARP-consistency. The Afriat-multiplier formulation is bilinear in the unknowns
 125 once $\tilde{x}$ is itself a decision variable; we instead use the multiplier-free ordinal characterization of
 126 Demuynck and Rehbeck [2023], under which every constraint is linear. At $T = 25$ and $G = 2$
 127 this is 600 binaries and 125 continuous variables per trace, solved as a single MILP on the HiGHS

backend [Huangfu and Hall, 2018]. The full formulation and the three method commitments are in Appendix B. All 85 projections were re-verified GARP-consistent by the Warshall-closure GARP check [Warshall, 1962], independently of the solver (MIP gap $\leq 8.1 \times 10^{-5}$ on every solve).

The objective is L_1 : minimal total absolute displacement between $\tilde{x}$ and x across goods and rounds (L_∞ is recorded alongside; L_2 would require an MIQP formulation). Weights are uniform, which keeps the program a pure MILP. The *dose* for a trace is its L_1 projection distance $\sum_t \sum_k |x_{t,k} - \tilde{x}_{t,k}|$ (Figure 1, step 5), comparable across sessions without further normalization because income is fixed throughout. It is the same style of graded, minimum-cost severity measure as Dean and Martin [2016], applied here to an LLM’s own choices and paired with the exogenous payoff of §3.2. Wherever a mean dose or mean Δ payoff is reported over a group of traces, it is averaged over every trace in that group, with 0 for any already-GARP-consistent trace. We compute a Cobb–Douglas demand whose expenditure shares are fitted to each trace’s observed data as an upper bound on the reported distance; it is not supplied to the solver (Appendix A, check 1). Because the GARP-consistent set is a union of polyhedra rather than a single convex set, we claim no distance-minimization guarantee analogous to a convex-cone projection [Wang et al., 2026].

3.2 The exogenous payoff

The payoff is fixed independently of the agent’s revealed choices. A utility function fit to the agent’s choices (such as the share-fitted Cobb–Douglas demand of §3.1) would be circular if used as the outcome measure, rewarding the agent for resembling its own revealed preferences. We use an equal-weight Cobb–Douglas valuation $U_{\text{exo}}(x) = x_A^{0.5} x_B^{0.5}$, fixed before any model was queried. It was never re-estimated from any agent’s choices. At budget line (p_t, I_t) the optimal bundle x_t^* is the closed-form Cobb–Douglas demand $x_{t,A}^* = 0.5I_t/p_{t,A}$, $x_{t,B}^* = 0.5I_t/p_{t,B}$, which depends only on prices and income. The payoff score is the efficiency ratio $\text{payoff}(x_t) = U_{\text{exo}}(x_t)/U_{\text{exo}}(x_t^*) \in (0, 1]$, computed identically for the raw and projected sequence at the same budget lines, so that $\Delta\text{payoff} = \text{payoff}(\tilde{x}) - \text{payoff}(x)$ is comparable across conditions, models, and repairs. This has the structure of a money-metric utility index, the standard welfare-comparison device over the same GARP/Afriat objects this paper’s method already relies on. Ouyang et al. [2024] give a comparable exogenous, real-world payoff (LLM-generated forecasts scored against firms’ realized future spending, finding a *non-monotonic* alignment-accuracy relationship), but lack a placebo of matched size to separate an alignment-specific effect from a generic fine-tuning effect; our design makes that separation.

We checked four failure modes of this design: a shared-data leak between payoff and projection, a mechanical link between GARP-consistency and payoff, a projection direction correlated with the payoff optimum, and the confound that larger-dose traces start with lower payoff and have more room to improve. All four are ruled out — the fourth, the strongest of them, leaves the dose coefficient significant once raw payoff is controlled for (Appendix A gives every check and number). A fifth concern, that this payoff’s concavity could reward displacement alone, motivates the control constructed next.

3.3 Null-operator controls

The payoff of §3.2 reduces to $\text{payoff}(s) = 2\sqrt{s(1-s)}$, where s is the expenditure share on good A. It is concave with a single interior maximum at $s = 0.5$, identical across every trace. Any operator that moves a bundle toward that fixed point raises that bundle’s payoff, provided the displacement does not overshoot the optimum, whether or not it restores GARP-consistency. We construct a *null operator* that shrinks every observed bundle in a GARP-violating trace toward the exogenous optimum x_t^* by a single shared factor $\lambda \in [0, 1]$, chosen so the null’s total L_1 displacement exactly matches the real projection’s dose. The null does not call the GARP check. It restores consistency in none of the 85 traces; it is scored by the payoff of §3.2.

Because that fixed target is identical across every trace, we build a second payoff that removes it: a per-trace Cobb–Douglas weight $\alpha_i \sim \text{Uniform}(0.05, 0.95)$, drawn independently for each of the 85 traces i and each of $K = 20$ replicate draws, read only at scoring time. We report two null constructions under this payoff: a *primary* null, unaware of α_i , and an *oracle* null that aims at the true per-trace optimum $x_{\alpha_i}^*$. The two are reported separately. The oracle null is an upper bound: it uses the per-trace α_i , which a repair algorithm does not observe.

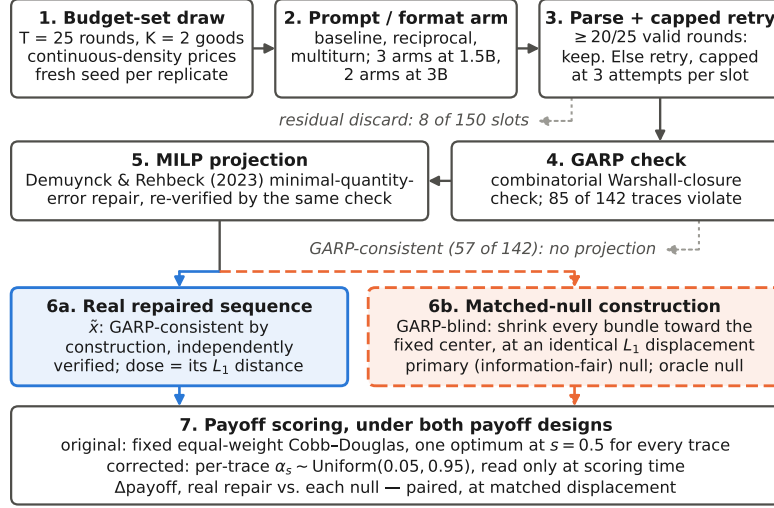


Figure 1: The pipeline, one replicate end to end: budget-set draw (1); prompt/format condition (2); parse (3); GARP check (4); and, for the traces that violate it, the minimal-quantity-error MILP of §3.1 (5). The real repaired sequence (6a, solid blue) and the GARP-blind null spending an identical L_1 displacement (6b, dashed orange) are scored (7) under both payoff designs.

180 4 Experimental design

181 4.1 Models and conditions

182 We use two locally hosted open-weight models: `qwen2.5:1.5b-instruct` at 1.5B and
 183 `llama3.2:3b-instruct` at 3B. The pilot found within-condition coherence variation at 1.5B and
 184 mean baseline CCEI 0.99 at 3B, so 1.5B carries the dose–response measurement and 3B serves as a
 185 low-headroom control. Models were run one at a time, all runs for one model completing before the
 186 next was loaded; concurrent residency of a third model family caused reproducible crashes. **Compute.**
 187 All inference ran locally on 4-bit-quantized weights, with no GPU and no commercial API calls. The
 188 full $N = 30$ design completed in ≈ 2.1 hours wall-clock, every MILP projection (§3.1) solving in
 189 under 5 seconds.

190 Three conditions at 1.5B, two at 3B. **Baseline.** Direct per-token-return framing, single turn, all 25
 191 rounds in one prompt and response. **Reciprocal.** The same single-turn format with an inverted-price
 192 framing manipulation, the strongest manipulation identified in the pilot. **Multi-turn** (1.5B only).
 193 Baseline framing, but each of the 25 rounds is a separate sequential call carrying the running conver-
 194 sation history, which isolates the elicitation-format mechanism from the price-framing mechanism.
 195 Wang et al. [2025a] collapse multi-turn to single-turn in the same domain and find that this drops
 196 CCEI by 0.241 for a same-family model at 7B; because our baseline is already single-turn, this
 197 condition tests the reverse. The 3B model is a low-headroom control, so this condition was run at
 198 1.5B only.

199 4.2 Budget sets and power

200 Each of $N = 30$ independent replicates per (model, condition) cell draws a fresh $T = 25$, $G = 2$
 201 budget set (prices $p_A, p_B \sim U[0.1, 1.0]$ i.i.d., rejected unless $\max(p_A, p_B) \geq 0.5$, budget exhaustion
 202 enforced), seeded independently to support between-condition replication. A *slot* is one such design
 203 position; each slot is run for up to three *attempts*, one session each (§4.3); a completed attempt
 204 yields one *trace*. Continuous, independently drawn prices with enforced budget exhaustion also
 205 avoid a pathology documented by Andrews [2026]: CCEI can read exactly 1.0 on GARP-violating
 206 data whenever a comparison lands on exact budget-line equality, common under grid-sampled or
 207 rounded prices. $N = 30$ gives power ≥ 0.999 against the pilot’s full framing-effect magnitude on
 208 the pass-rate metric and ≥ 0.80 down to an effect attenuated to $\sim 60\%$ of that magnitude (pass rate
 209 $0.76 \rightarrow 0.40$, vs. the pilot’s $0.76 \rightarrow 0.17$), but is underpowered for a CCEI-scale effect of the pilot’s

Table 2: Headline results per (model, condition) cell. GARP pass and CCEI are computed on kept traces only; dose and Δpayoff are averaged over all kept traces, with 0 for any already-GARP-consistent trace, and are under the original payoff (§3.2).

Model	Condition	n kept	mean CCEI	GARP pass	mean dose (L_1)	mean Δpayoff
llama3.2:3b	baseline	30	0.9900	0.73	5.01	+0.0018
llama3.2:3b	reciprocal	28	0.9713	0.39	6.27	+0.0016
qwen2.5:1.5b	baseline	30	0.9522	0.40	16.68	+0.0090
qwen2.5:1.5b	multi-turn	30	0.9540	0.10	14.54	+0.0083
qwen2.5:1.5b	reciprocal	24	0.9413	0.38	12.02	+0.0065

210 magnitude at 1.5B (minimum detectable ≈ 0.11). Every replicate’s Bronars power [Bronars, 1987] is
 211 recorded; all 150 draws held Bronars power ≥ 0.998 .

212 4.3 Discard selection and the retry protocol

213 The pilot found that under reciprocal framing at 1.5B, 52% of sessions (13 of 25) failed the required
 214 output format and were silently discarded under the pilot’s handling — standard practice, whose
 215 consequences this paper measures as an instrument-validity claim. The main experiment implements
 216 a capped retry protocol: a session failing the $\geq 20/25$ -valid-rounds threshold is retried up to three
 217 times with a new random seed; a slot still failing after three attempts is a residual discard, excluded
 218 from the main CCEI/GARP analysis but scored separately for the audit in Table 3 (§5.2). Three of
 219 the 30 qwen reciprocal slots returned zero valid rounds on all three attempts, yielding no price or
 220 quantity data.

221 5 Results

222 We ran 150 slots (llama3.2:3b in 2 conditions, qwen2.5:1.5b in 3, 30 replicates each) and made 187
 223 attempts. Of these, 142 slots kept a usable trace and 8 were residual discards after three attempts.
 224 Every cell was run to its full 30 slots.

225 5.1 Dose–response and the null-operator control

226 Across the 85 GARP-violating traces (60 at 1.5B, 25 at 3B): Spearman $\rho = 0.729$ ($p < 0.0001$),
 227 Pearson $r = 0.821$ ($p < 0.0001$), mean $\Delta\text{payoff} + 0.0091$ (one-sample $t = 5.41$, $p < 0.00001$),
 228 with 70 of 85 traces (82%) improving after repair and 15 (18%) worsening. The relationship holds
 229 at each scale separately: $\rho = 0.756$ ($p < 0.0001$, $n = 60$) at 1.5B and $\rho = 0.614$ ($p = 0.0011$,
 230 $n = 25$) at 3B. We do not compare its magnitude across scale. We tested the mechanical confound
 231 that larger-dose traces might start from worse raw payoff and therefore have more room to improve.
 232 The confound is real (dose vs. raw payoff, Pearson $r = -0.41$, $p < 0.0001$) but does not explain
 233 the relationship away. The partial correlation of dose against Δpayoff controlling for raw payoff is
 234 $r = 0.784$ ($p = 7 \times 10^{-19}$). An OLS regression of Δpayoff on dose and raw payoff jointly finds
 235 dose significant ($t = 11.4$, $p < 10^{-16}$), with raw payoff’s marginal contribution not distinguishable
 236 from zero ($p = 0.18$) once dose is included (Appendix A).

237 The raw-payoff confound bears on the *severity* of the raw violation. It leaves open whether restoring
 238 GARP-consistency or the displacement does the work. The null operator (§3.3) spends the identical
 239 L_1 budget as the real repair and shrinks every bundle toward the exogenous payoff’s fixed optimum,
 240 using no GARP information.

241 **Experiment 1 (original payoff).** The null operator outperforms the real repair by 2.4 $\times$: mean
 242 $\Delta\text{payoff}_{\text{null}} = 0.0220$ vs. mean $\Delta\text{payoff}_{\text{real}} = 0.0091$, Wilcoxon signed-rank $p = 3.98 \times 10^{-10}$,
 243 with the null winning 69 of 85 traces (81%; Fig. 2A). This holds at both model scales, and the
 244 partial correlation of dose against $\Delta\text{payoff}_{\text{real}}$, controlling for the null’s mechanically predicted gain,
 245 remains positive and significant at both scales separately ($r = 0.574$ pooled, $p = 1.13 \times 10^{-8}$;
 246 per-model breakdown in Table 5, Appendix A), so dose retains information beyond what the null
 247 predicts.

248 **Experiment 2 (corrected payoff).** The original payoff’s exploit is a single fixed target ($s = 0.5$)
 249 identical across every trace. Experiment 2 draws an independent Cobb–Douglas weight $\alpha_i \sim$

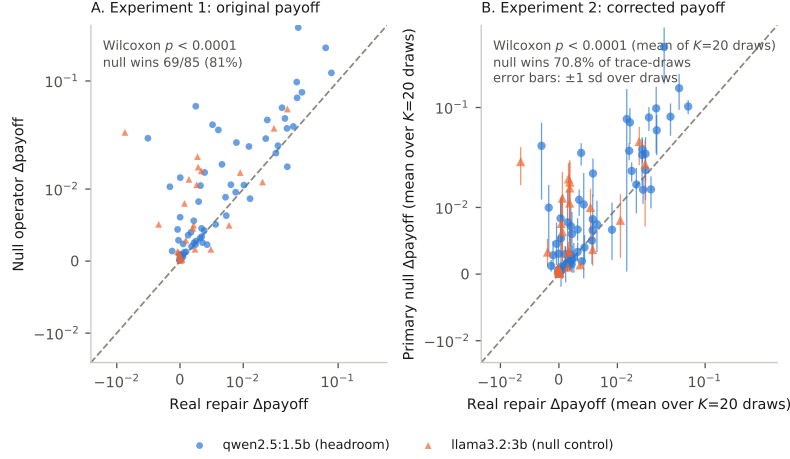


Figure 2: Real repair’s Δ payoff (x-axis) against the dose-matched null operator’s Δ payoff (y-axis) on symmetric-log axes, one point per GARP-violating trace ($n = 85$; circle/blue: qwen2.5:1.5b, triangle/orange: llama3.2:3b). Dashed line is $y = x$; points above it have higher null Δ payoff than real. (A) Experiment 1, original payoff; (B) Experiment 2, corrected payoff, with each trace’s primary-null Δ payoff averaged over $K = 20$ draws of the random target (± 1 SD error bars).

Uniform(0.05, 0.95) per trace, over $K = 20$ independent draws (§3.3). Experiment 2 reproduces Experiment 1: the primary null still outperforms the real repair, by $3.0\times$ (mean $\Delta\text{payoff}_{\text{null}} = 0.02167$ vs. mean $\Delta\text{payoff}_{\text{real}} = 0.00723$, Wilcoxon $p = 1.24 \times 10^{-5}$, win rate 70.8%), significant in 20 of 20 draws and in both models separately. The oracle null, privileged to know the true per-trace target, wins 80.4% of traces (mean $\Delta\text{payoff}_{\text{null}} = 0.02224$, $p = 1.35 \times 10^{-9}$, significant in 20 of 20 draws, also for both models separately). The partial correlation of dose with $\Delta\text{payoff}_{\text{real}}$, controlling for the null, attenuates from Experiment 1’s $r = 0.574$ to a mean $r = 0.37$ across the 20 draws (SD 0.227 primary-null, 0.223 oracle-null), significant in 14 of 20 (primary) and 15 of 20 (oracle) draws; the attenuation and the weaker partial correlation with the real repair’s own gain both hold per model, more sharply for qwen2.5:1.5b (Table 5).

Two payoff designs. One is the paper’s original; the second was built to remove the first one’s exploitable structure. Both return the same negative paired comparison. A third payoff, selected after seeing that the second was also negative, would be payoff-shopping (§6).

Trace extremity and the null’s advantage. Trace extremity (1 – raw payoff under the original fixed payoff, a measure of the raw bundles independent of which payoff later scores them) predicts the null’s advantage in both experiments (Experiment 1: Pearson $r = 0.679$; Experiment 2 primary null: mean $r = 0.631$ across 20 draws), and predicts the real repair’s own gain less strongly in both. This pattern does not separate “centering” from a more general account on which moving an extreme trace toward the interior helps regardless of destination, since Experiment 2’s primary null is mechanically the same fixed-center operator as Experiment 1’s. Experiment 2’s oracle null, which targets a per-trace-varying point, shows the same extremity-advantage to within 0.01 (mean $r = 0.624$ against 0.631). We treat the “escape a bad start” reading as a hypothesis for the discussion; we do not test it. Figure 3 illustrates it on a single trace. On the largest-dose trace the repair changes 5 of 21 rounds, spending 42% of its L_1 budget on round 3 alone ($s = 0.11 \rightarrow 0.65$, payoff $0.626 \rightarrow 0.953$) and leaving the other 16 at a near-corner $s = 0.99$, while the null spreads the same total displacement over all 21 rounds.

The hypothesis that restoring GARP-consistency, beyond displacement toward the interior of the budget line, buys more exogenous payoff (C1) is not supported under either payoff design. The raw dose–response relationship holds; it is not evidence for the specific mechanism C1 claims.

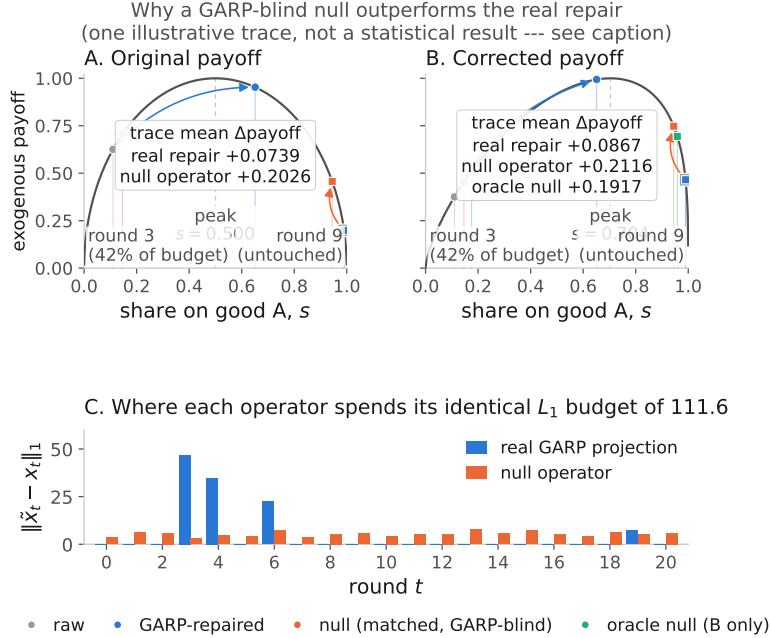


Figure 3: The largest-dose trace (qwen2.5:1.5b, reciprocal, replicate 12; $L_1 = 111.64$, $T = 21$ rounds). With $G = 2$, payoff is a function of expenditure share s . **(A)** The original payoff $2\sqrt{s(1-s)}$, peak at $s = 0.5$; **(B)** the corrected payoff at this trace’s draw $\alpha_i = 0.704$; **(C)** per-round s across all 21 rounds for the raw trace, the real repair, and the null operator.

5.2 Reciprocal framing and discard selection

At 3B, reciprocal framing replicates the pilot’s finding on an independently drawn budget set: GARP pass rate drops from 0.73 (22/30) to 0.39 (11/28), $p = 0.0089$ (Benjamini–Hochberg $p_{BH} = 0.011$, Holm $p_{Holm} = 0.062$); CCEI does not reach significance (0.9900 vs. 0.9713, t -test $p = 0.0508$). CCEI compresses near 1, as the pilot found. Here, GARP pass rate separated baseline from reciprocal and CCEI did not.

At 1.5B, the reciprocal-framing effect does not survive correction for discard-selection. First-attempt discard under reciprocal framing was 43.3% (13/30), against the pilot’s unretried 52%, falling to a residual 20.0% (6/30) after the retry protocol; it produces more discards than any other condition, including the other 1.5B manipulation (multi-turn, 0%). Once the surviving 24 sessions are compared, GARP pass rate (0.40 vs. 0.38, $p = 0.85$) and CCEI (0.9522 vs. 0.9413, $p = 0.73$) are indistinguishable between baseline and reciprocal framing; neither experiment detects a CCEI shift, as the pilot’s +0.0169 ($p = 0.66$) and the main experiment’s -0.0109 ($p = 0.73$) are two statistically indistinguishable nulls.

Retry-rescued sessions differ from first-attempt sessions: mean CCEI 0.9651 against 0.9315, but GARP pass rate 0.1429 against 0.4706. The residual-discard group has the lowest mean CCEI of the three (0.8821), though its GARP pass rate is measured on only 3 slots. The residual sample is too small ($n = 6$, 3 measurable) to be conclusive, but the pattern is consistent with the pilot’s selection concern surviving the retry correction at reduced magnitude. At 3B, the first-attempt and retry-rescued groups differ by 0.0118 in CCEI and 0.143 in GARP pass rate, so the selection concern shows up more sharply at 1.5B than at 3B. Table 3 breaks the discards down by attempt.

5.3 Multi-turn elicitation format

Replacing the single prompt with 25 sequential calls drops the GARP pass rate from 40% to 10% ($p = 0.0073$) while CCEI does not move (0.9522 vs. 0.9540, $p = 0.91$; Table 4), with zero discards in either condition. Reciprocal framing produced no detectable drop at 1.5B, so the multi-turn format is the only 1.5B manipulation with a measured pass-rate effect. The drop survives Benjamini–Hochberg

Table 3: Discard-outcome breakdown for all 60 reciprocal-framing sessions (30 per model), by attempt group: kept on the first attempt, rescued by a later retry, or still discarded after three attempts (§4.3). GARP pass, CCEI and mean dose (L_1) follow the conventions of §3.1 and Table 2. *Computed over the 3 of 6 qwen residual-discard slots with any valid rounds.

Model	Attempt group	n	mean CCEI	GARP pass	mean dose (L_1)
llama3.2:3b	first-attempt success	21	0.9742	0.4286	5.086
llama3.2:3b	retry-rescued	7	0.9624	0.2857	9.839
llama3.2:3b	residual discard	2	1.0000	1.0000	0.000
qwen2.5:1.5b	first-attempt success	17	0.9315	0.4706	13.847
qwen2.5:1.5b	retry-rescued	7	0.9651	0.1429	7.600
qwen2.5:1.5b	residual discard	6	0.8821*	0.6667*	24.196*

correction ($p_{\text{BH}} = 0.0094$) but not Holm ($p_{\text{Holm}} = 0.058$). Together with the 3B reciprocal-framing result (§5.2), GARP pass rate again separates the conditions and CCEI does not. Wang et al. [2025a] report the opposite: in the same domain, on the same model family at 7B (Qwen2.5-7B), single-turn elicitation was more coherence-degrading than multi-turn; here, at 1.5B, the reverse holds. We have no explanation for the reversal.

Table 4: 1.5B (qwen2.5:1.5b), baseline vs. multi-turn elicitation format (25 sequential calls), $n = 30$ per condition. GARP pass rate is reported as count/ n with 95% Wilson score confidence intervals (Pearson χ^2 test, $p = 0.0073$); mean CCEI is reported with 95% t -based confidence intervals ($df = 29$; two-sample t -test, $p = 0.91$).

Condition	GARP pass	95% CI (Wilson)	mean CCEI [95% CI (t)]
baseline	12/30 (40.0%)	[24.6, 57.7]%	0.9522 [0.9253, 0.9791]
multi-turn (25 seq. calls)	3/30 (10.0%)	[3.5, 25.6]%	0.9540 [0.9350, 0.9729]

6 Limitations

CCEI is underpowered at 1.5B. Baseline WSCV (Within-Subject Coefficient of Variation) 14.8% is comparable to published human test–retest noise [Nitsch et al., 2022], and $N = 30$ is underpowered for a CCEI-scale effect of the pilot’s magnitude (minimum detectable $2.5\times$ the pilot’s estimate; a CCEI-power-matched design would need $N = 111\text{--}161$, which we did not run). The pass-rate metric is powered ≥ 0.999 throughout, and every CCEI result carries its own confidence interval.

Single run per condition. Each (model, condition) cell is one run of the stated protocol: we report no between-run variance on the headline statistics, only the within-run replicate variance already in the reported confidence intervals and p -values.

One scale point per model. The two-model design isolates the identification strategy from a scale confound but does not trace a dose–response relationship *across* scale: in our model-selection sweep (§4), coherence headroom and measurement reliability did not coexist in any model tested, and a third model family was excluded for compatibility reasons.

Two exogenous payoffs. The first payoff’s single, globally identical optimum is exploitable by a GARP-blind operator; the second (§3.2, §3.3) removes that property and reproduces the same verdict. A third payoff, chosen after seeing the second was also negative, would be payoff-shopping — the researcher-degree-of-freedom problem an exogenous payoff rules out (§5.1).

Consistent with three prior negative results. Three independently published repair attempts moved the wrong way (§2); our null-operator result (§5.1) is a fourth. Separately, our reciprocal-framing manipulation at 1.5B (§5.2) failed to induce the coherence variation it targeted. That leaves C2 open; it is not evidence the relationship is negative in general.

7 Broader Impacts

This result argues against imposing GARP repair on AI economic agents as a routine intervention without a matched control: under two payoff designs, a GARP-blind operator outperforms the real

334 repair (§5.1), so a practitioner relying on a raw within-scale dose–response correlation alone would
 335 act on the confounded estimate this paper’s control was built to separate. Correcting discard selection
 336 removes the reciprocal-framing manipulation’s apparent coherence effect (§5.2); the design measures
 337 whether repair helps, and here it does not. This instrument-validity finding (§4.3) is not specific to
 338 our instrument: any consistency instrument that drops disruption-caused failures without recording
 339 them will underestimate the disrupting manipulation. We see no elevated misuse risk specific to this
 340 work: synthetic tasks, open-weight models, no new dataset, no human subjects.

341 8 Conclusion

342 Restoring GARP-consistency in an LLM agent’s own realized choices does not buy more exogenous
 343 payoff than displacement alone: at both model scales and under two payoff designs, a dose-matched,
 344 GARP-blind null operator outperforms the real repair on the same payoff. C1 is not supported.
 345 Trace extremity predicts the null’s advantage in both experiments, a candidate explanation that does
 346 not establish the mechanism. C2, the coherence–competence question, remains open — this is the
 347 first attempt we know of to measure it without confounding coherence with capacity, a preference-
 348 judgment outcome, or displacement magnitude — and a practitioner evaluating coherence repair
 349 should include a dose-matched null operator before reading a dose–response correlation as evidence
 350 of benefit. Two further findings stand on their own: a framing manipulation’s apparent effect was
 351 a discard-selection artifact (C3), and an elicitation-format manipulation moved GARP pass rate
 352 opposite to Wang et al. [2025a]’s finding in the same domain and model family at a larger scale.

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416 A Payoff exogeneity audit

- 417 **(1) Payoff inputs.** The payoff is computed from (p_t, I_t) by a routine that reads no projection output;
418 the optimal bundle x_t^* is a function of (p_t, I_t) alone and never references x_t or $\tilde{x}_t$. The share-fitted
419 Cobb–Douglas demand of §3.1 is used only to log an upper bound on the reported distance. It is
420 never passed to HiGHS as an incumbent solution, for which the solver exposes no interface.
- 421 **(2) Independence of the payoff optimum from revealed preference.** Raw (pre-repair) payoff does
422 not differ between the 57 already-GARP-consistent traces and the 85 violating traces (0.7872 vs.

0.7899; Welch $t = -0.07$, $p = 0.94$). No population-level link between GARP-consistency and payoff is detectable.

(3) Trace-level mechanism check. Five traces spanning the dose range (0.17 to 111.64) were inspected by hand, comparing the direction of the actual projection movement ($\tilde{x} - x$) against the direction toward the exogenous optimum ($x^* - x$) via cosine similarity; every quantity plotted in Figure 3 was recomputed by re-solving the projection MILP. Similarities were low (0.16–0.31) on the four traces where payoff improved after repair, and negative (−0.35) on the single trace where payoff worsened. On these five traces the sign of the cosine similarity matched the sign of the payoff change, and the cosine of the projection step against ($x^* - x$) appears nowhere in the projection objective. This is evidence against the projection acting as a disguised optimization toward the payoff target.

(4) Severity confound. Larger-dose traces start from worse raw payoff (Pearson $r = -0.411$, $p = 9.4 \times 10^{-5}$), and worse-starting traces improve more ($r = -0.415$, $p = 8.0 \times 10^{-5}$, between raw payoff and Δpayoff). These are two distinct correlations that round to the same −0.41; neither is a restatement of the other. This is a ceiling-effect confound. The dose–response relationship survives it: the partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$). The unconditional correlation is $r = 0.821$. In a joint OLS regression the dose coefficient is significant ($t = 11.43$, $p < 10^{-16}$) while raw payoff’s coefficient is not ($t = -1.36$, $p = 0.18$).

No part of the audit found a channel by which the projection could influence payoff other than through (p_t, I_t) .

(5) Per-model breakdown of the null-operator comparison. Table 5 gives the numbers pooled in §5.1, split by model.

Table 5: Per-model breakdown of the null-operator comparison (§5.1). Experiment 1 uses the original payoff. Experiment 2 uses the corrected payoff, aggregated over $K = 20$ draws (mean values; “sig.” counts are draws with Wilcoxon $p < 0.05$ out of 20). Partial r is the partial correlation of dose against $\Delta\text{payoff}_{\text{real}}$ controlling for the null’s predicted gain (mean and, for Experiment 2, its significance count).

Experiment	Model	n	$\Delta\text{payoff}_{\text{real}}$	$\Delta\text{payoff}_{\text{null}}$	partial r
1, real null	llama3.2:3b	25	0.0039	0.0105 ($p = 0.0010$)	0.750
1, real null	qwen2.5:1.5b	60	0.0113	0.0267 ($p = 5.5 \times 10^{-8}$)	0.567
2, primary null	llama3.2:3b	25	0.00276	0.00906 (16/20 sig.)	0.553 (18/20 sig.)
2, primary null	qwen2.5:1.5b	60	0.00910	0.02693 (20/20 sig.)	0.341 (12/20 sig.)
2, oracle null	llama3.2:3b	25	0.00276	0.00985 (20/20 sig.)	—
2, oracle null	qwen2.5:1.5b	60	0.00910	0.02741 (20/20 sig.)	—

B MILP formulation and implementation

Formulation. We use the multiplier-free ordinal characterization of Demuynck and Rehbeck [2023], under which every constraint is linear jointly in the perturbed bundles, a set of ordinal utility levels $u_t \in [0, 1]$, and binary comparison indicators $U_{t,v} \in \{0, 1\}$ for $t \neq v$. The formulation is solved as a single MILP, with no outer search over preference orderings. Afriat-multiplier parameterizations are bilinear once $\tilde{x}$ is a decision variable, and remain so under any fixed preference ordering, because a price-weighted multiplier term stays a product of two unknowns.

Budget exhaustion. Budget exhaustion is imposed as an equality, $p_t \cdot \tilde{x}_t = I_t$ for every observation, with income fixed at 100 throughout. The big- M constant in the ordering constraints is therefore fixed a priori: any $M > 100$ is valid, and it needs no tuning.

Strict-preference margin. The GARP-consistent set is not closed, so an unregularized projection can read a distance of exactly zero on violating data. A fixed strict-preference margin $\gamma > 0$, set to $10^{-4} \cdot I = 10^{-2}$, converts an unattained infimum into an attained minimum. Re-solving at $\gamma \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ on a dose-stratified sample of 17 of the 85 GARP-violating traces (spanning the full observed dose range, 0.17–111.64) changes the reported L_1 distance by at most 16.8% relative to the operating-point value, and by under 1.7% on all but the single lowest-dose

461 trace; independent GARP-consistency re-verification fails on 3 of the 17 traces at the smallest margin
 462 tested ($\gamma = 10^{-6}$), so we do not recommend γ below 10^{-4} .

463 **Independent verification.** Every returned $\tilde{x}$ is verified by the Warshall-closure [Warshall, 1962]
 464 GARP check of §3.1. All 85 GARP-violating traces’ projections were re-verified GARP-consistent;
 465 MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

466 **Objective.** $\min \sum_t \sum_k w_{t,k} d_{t,k}$ subject to $\tilde{x}_{t,k} - x_{t,k} \leq d_{t,k}$ and $x_{t,k} - \tilde{x}_{t,k} \leq d_{t,k}$, with uniform
 467 weights $w = 1$.

468 B.1 Scope of the guarantee

469 The closest published analogue is a proved-optimal projection onto a convex monotone cone [Wang
 470 et al., 2026]. There, non-expansiveness licenses a guarantee that the edit lands weakly closer to
 471 ground truth, because the cone is fixed by an ordering supplied as input. The GARP-consistent set is
 472 a union of polyhedra rather than a single convex set, and a GARP repair must itself decide which
 473 revealed-preference comparisons to give up. That decision is absorbed into the binary comparison
 474 indicators $U_{t,v}$ rather than eliminated, so we claim no distance-minimization guarantee.

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591 (prompts, seeds, the parser/validity rule, and raw per-trace logs for the main experiment, the
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619 Answer: [Yes]

620 Justification: §4 specifies model identities, all three conditions’ exact framing, the budget-set
621 generator and its rejection rule, seed scheme, replicate count and power calculation, and the
622 discard/retry protocol.

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